

The Challenge of Manufacturing

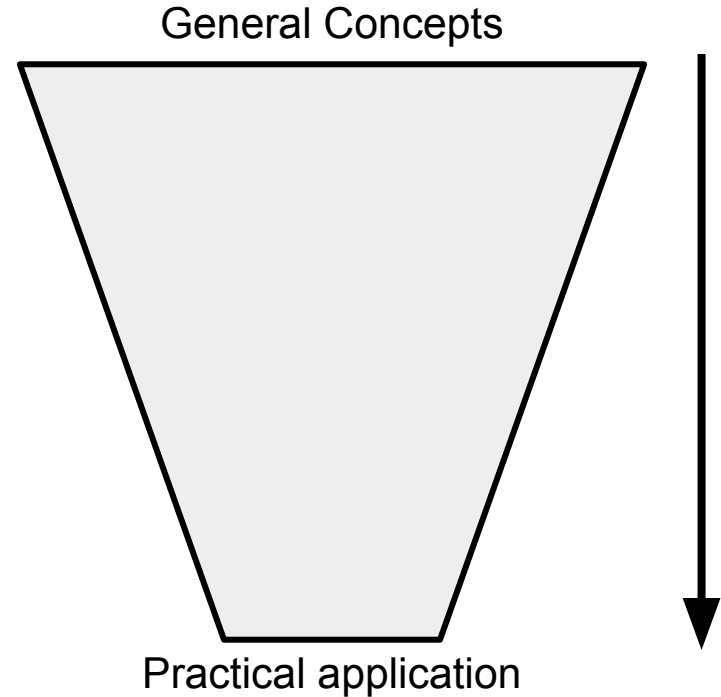
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Overview

- The drawing of As
- Numbers vs data
- What is variation?
- Specifications and economic loss function
- Inspection
- The definition of quality
- The Taguchi loss function
- The process principle
- Two types of variation
- An example



Is it always the case that
 $2 + 2 = 4$?

When does
 $2 + 2 \neq 4$?

Numbers

Represent the mathematical plane
NOT generated a process
Values are **fixed**

Data

Represent the real world
Generated by a process
Values **vary**

What is variation?

The degree to which the sequential outputs of a process or system differ.

Take the task of drawing a sequence of capital
letter As



We are limited in doing what we want to do; that to do what we set out to do, even in so simple a thing as making a's that are alike, requires almost infinite knowledge compared with that which we now possess. It follows, therefore, since we are thus willing to accept as axiomatic that we cannot do what we want to do and cannot hope to understand why we cannot, that we must also accept as axiomatic that a controlled quality will not be a constant quality. Instead, a controlled quality will be a variable quality.

— Walter A. Shewhart
Economic Control of Quality of Manufactured Product

Craft production

Low volume

Manual

Skilled craftsmen

Not interchangeable parts

Mass production

High volume

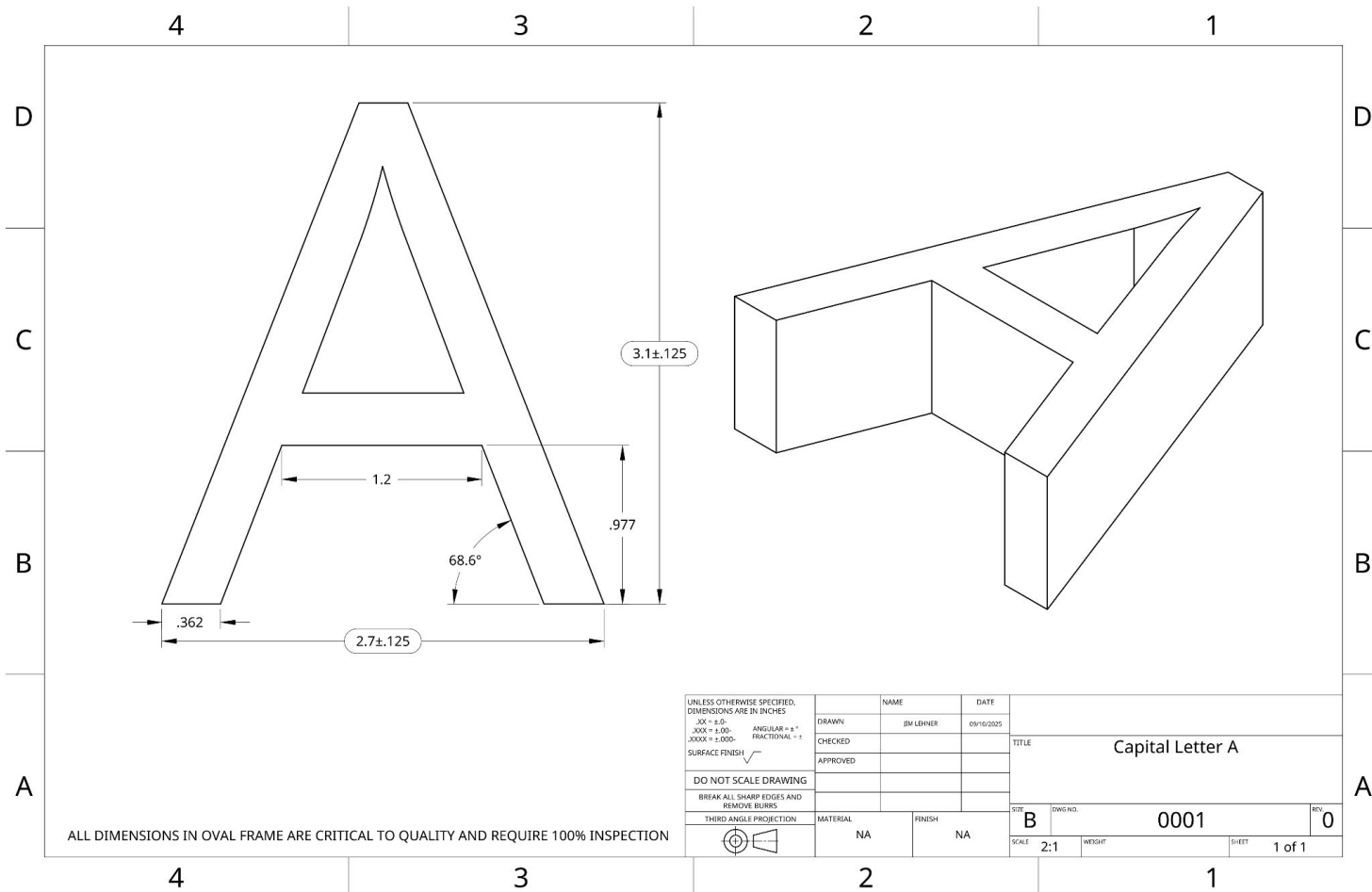
Automated

Unskilled labor

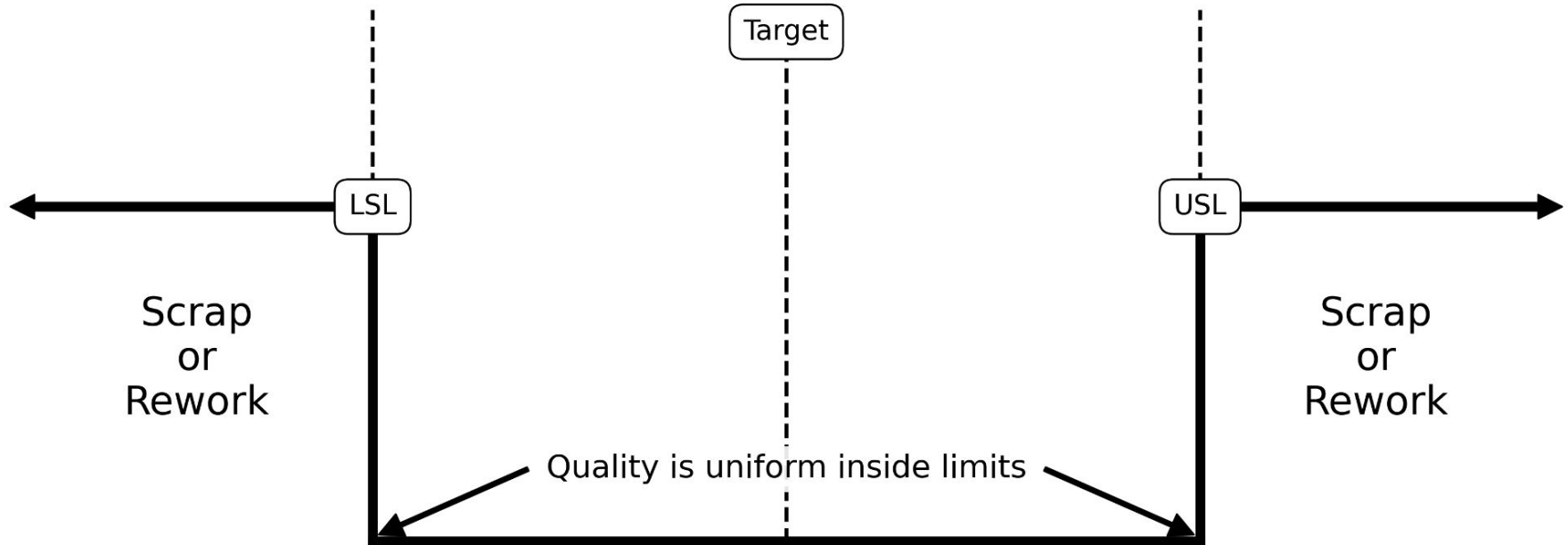
Interchangeable parts

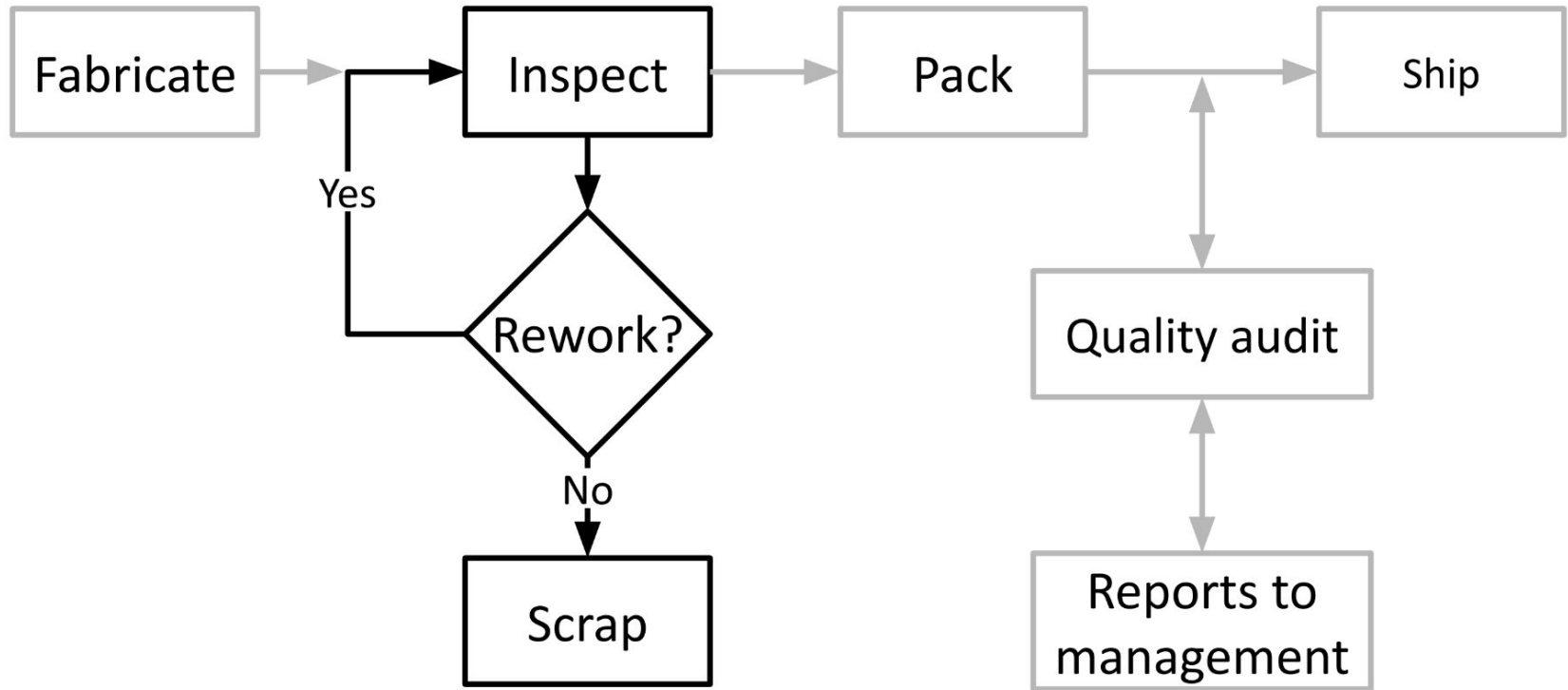
What are specifications?

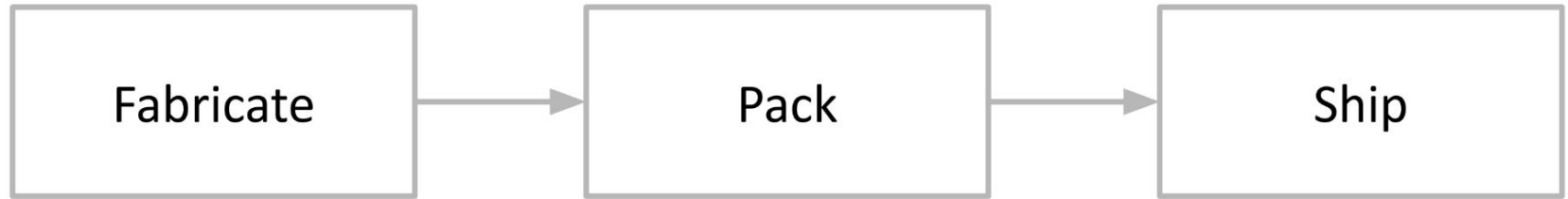
Specifications define the extent
to which parts can differ
(and still satisfy the customer need).



Rectangular loss function







What is the definition of quality?

On-target with minimum variance.

World-class quality can be approximated with a quadratic loss function of the form:

$$L(x) = K (x - T)^2$$

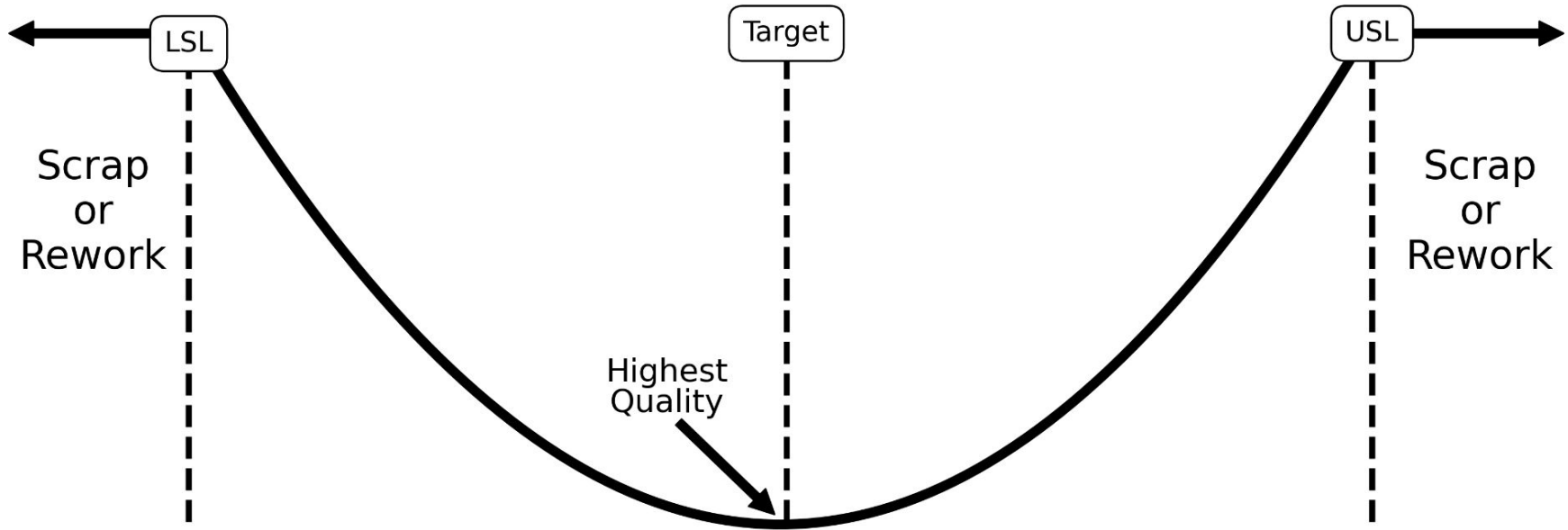
$L(x)$ is the loss function

K is a numeric constant

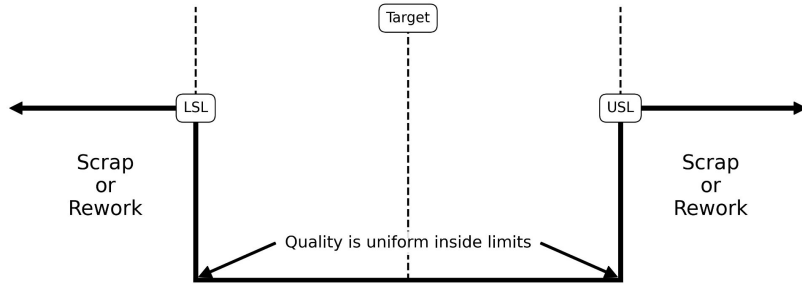
x is the value of the performance or quality characteristic

T is the target value

Taguchi loss function



Square loss function



Low fidelity model of loss
Loss occurs **abruptly**
Promotes **reactionary efforts**

Taguchi loss function



High fidelity model of loss
Loss occurs **incrementally**
Promotes **continual improvement**

The process principle:

The sequential outputs of a process are subject to some degree of difference, whether we can measure that difference or not.

Common causes

Produce routine variation

Many in number

Small effect on process behavior

Predictable and **consistent**

Controlled

Assignable causes

Produce exceptional variation

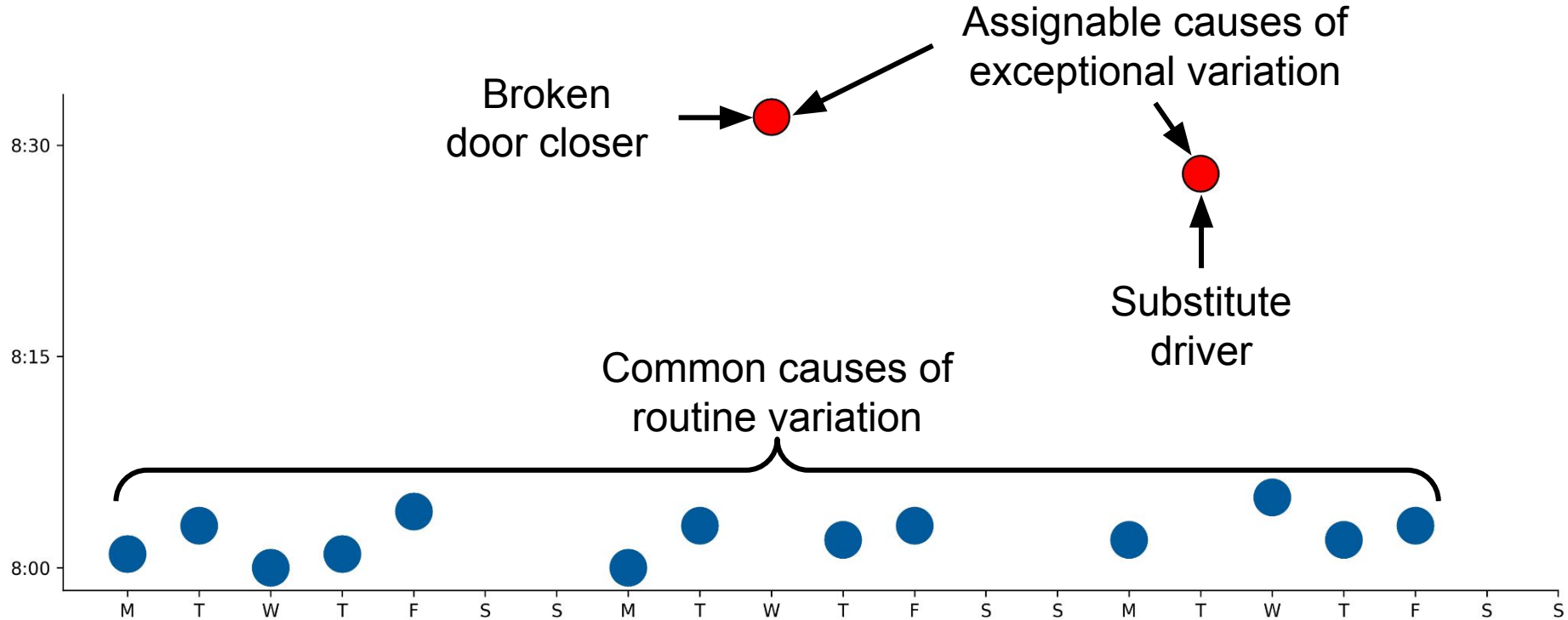
Few in number

Large effect on process behavior

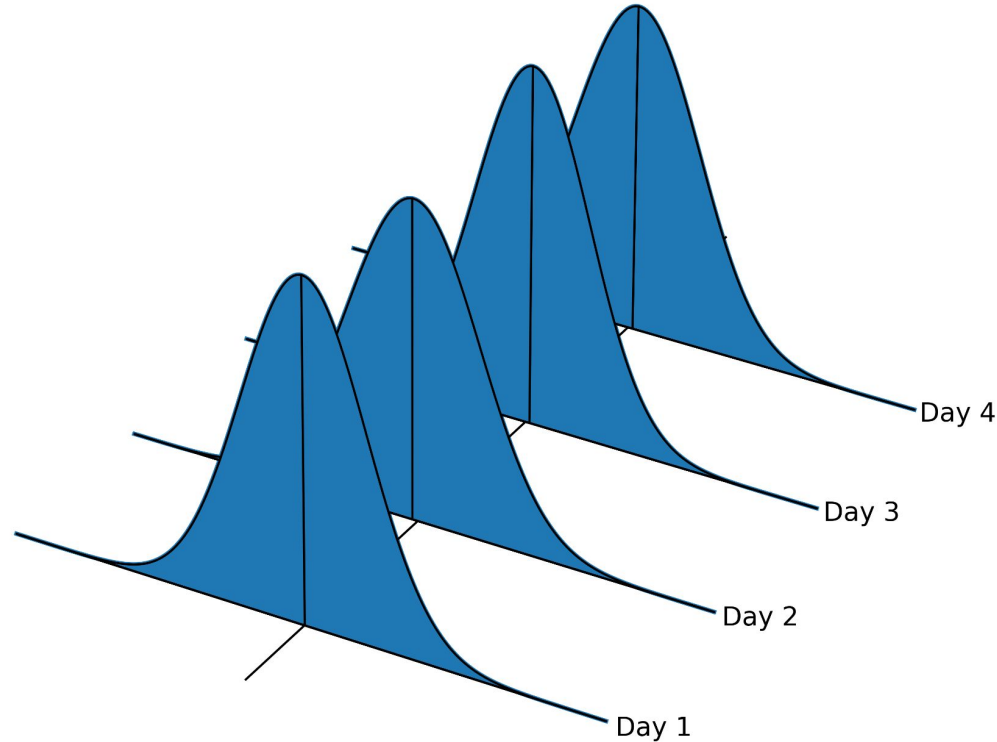
Unpredictable and **inconsistent**

Uncontrolled

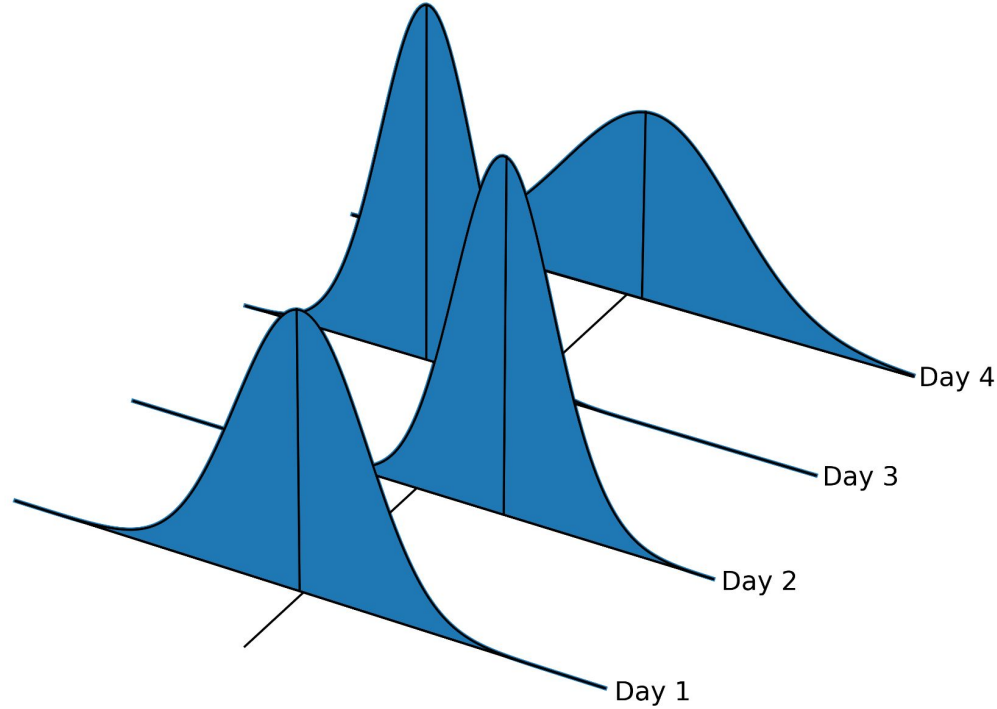
Arrival of the school bus



What can be said about these distributions?

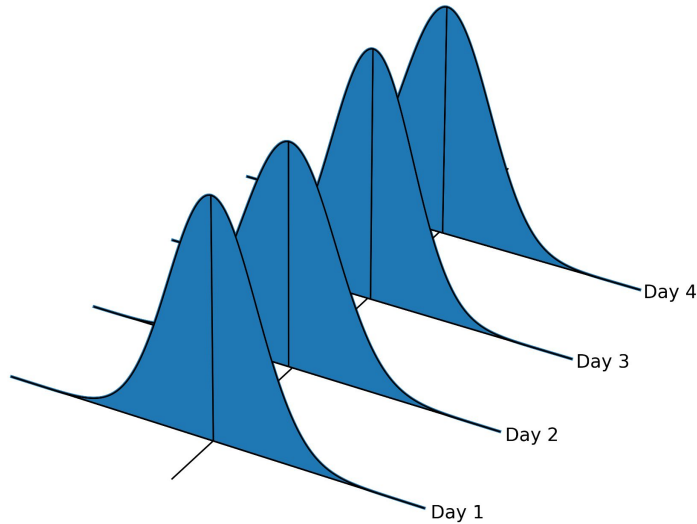


What can be said about these distributions?

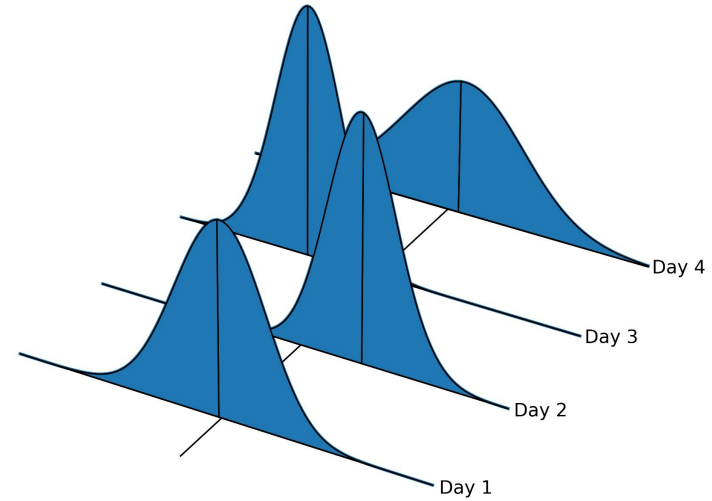


How do we know if a process is predictable or unpredictable?

Predictable process



Unpredictable process



A practical example

204 electrical resistance measurements

5,045	4,635	4,700	4,650	4,640	3,940	4,570	4,560	4,450	4,500	5,075	4,500
4,350	5,100	4,600	4,170	4,335	3,700	4,570	3,075	4,450	4,770	4,925	4,850
4,350	5,450	4,110	4,255	5,000	3,650	4,855	2,965	4,850	5,150	5,075	4,930
3,975	4,635	4,410	4,170	4,615	4,445	4,160	4,080	4,450	4,850	4,925	4,700
4,290	4,720	4,180	4,375	4,215	4,000	4,325	4,080	3,635	4,700	5,250	4,890
4,430	4,810	4,790	4,175	4,275	4,845	4,125	4,425	3,635	5,000	4,915	4,625
4,485	4,565	4,790	4,550	4,275	5,000	4,100	4,300	3,635	5,000	5,600	4,425
4,285	4,410	4,340	4,450	5,000	4,560	4,340	4,430	3,900	5,000	5,075	4,135
3,980	4,065	4,895	2,855	4,615	4,700	4,575	4,840	4,340	4,700	4,450	4,190
3,925	4,565	5,750	2,920	4,735	4,310	3,875	4,840	4,340	4,500	4,215	4,080
3,645	5,190	4,740	4,375	4,215	4,310	4,050	4,310	3,665	4,840	4,325	3,690
3,760	4,725	5,000	4,375	4,700	5,000	4,050	4,185	3,775	5,075	4,665	5,050
3,300	4,640	4,895	4,355	4,700	4,575	4,685	4,570	5,000	5,000	4,615	4,625
3,685	4,640	4,255	4,090	4,700	4,700	4,685	4,700	4,850	4,770	4,615	5,150
3,463	4,895	4,170	5,000	4,700	4,430	4,430	4,440	4,775	4,570	4,500	5,250
5,200	4,790	3,850	4,335	4,095	4,850	4,300	4,850	4,500	4,925	4,765	5,000
5,100	4,845	4,445	5,000	4,095	4,850	4,690	4,125	4,770	4,775	4,500	5,000



Given the summary statistics, what next?

5,045	4,635	4,700	4,650	4,640	3,940	4,570	4,560	4,450	4,500	5,075	4,500
4,350	5,100	4,600	4,170	4,335	3,700	4,570	3,075	4,450	4,770	4,925	4,850
4,350	5,450	4,110	4,255	5,000	3,650	4,855	2,965	4,850	5,150	5,075	4,930
3,975	4,635	4,410	4,170	4,615	4,445	4,160	4,080	4,450	4,850	4,925	4,700
4,290	4,720	4,180	4,375	4,215	4,000	4,325	4,080	3,635	4,700	5,250	4,890
4,430	4,810	4,790	4,175	4,275	4,845	4,125	4,425	3,635	5,000	4,915	4,625
4,485	4,565	4,790	4,550	4,275	5,000	4,100	4,300	3,635	5,000	5,600	4,425
4,285	4,410	4,340	4,450	5,000	4,560	4,340	4,430	3,900	5,000	5,075	4,135
3,980	4,065	4,895	2,855	4,615	4,700	4,575	4,840	4,340	4,700	4,450	4,190
3,925	4,565	5,750	2,920	4,735	4,310	3,875	4,840	4,340	4,500	4,215	4,080
3,645	5,190	4,740	4,375	4,215	4,310	4,050	4,310	3,665	4,840	4,325	3,690
3,760	4,725	5,000	4,375	4,700	5,000	4,050	4,185	3,775	5,075	4,665	5,050
3,300	4,640	4,895	4,355	4,700	4,575	4,685	4,570	5,000	5,000	4,615	4,625
3,685	4,640	4,255	4,090	4,700	4,700	4,685	4,700	4,850	4,770	4,615	5,150
3,463	4,895	4,170	5,000	4,700	4,430	4,430	4,440	4,775	4,570	4,500	5,250
5,200	4,790	3,850	4,335	4,095	4,850	4,300	4,850	4,500	4,925	4,765	5,000
5,100	4,845	4,445	5,000	4,095	4,850	4,690	4,125	4,770	4,775	4,500	5,000

Mean:
4498.2

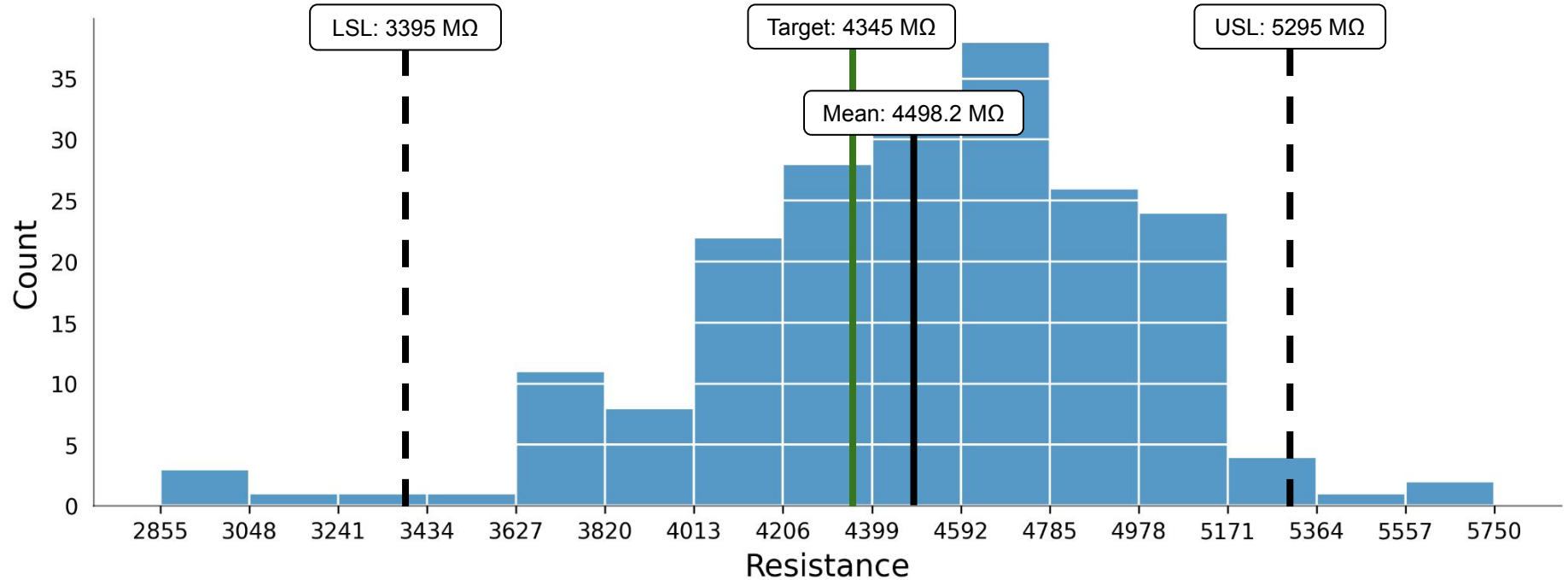
Median:
4565

Range:
2,895

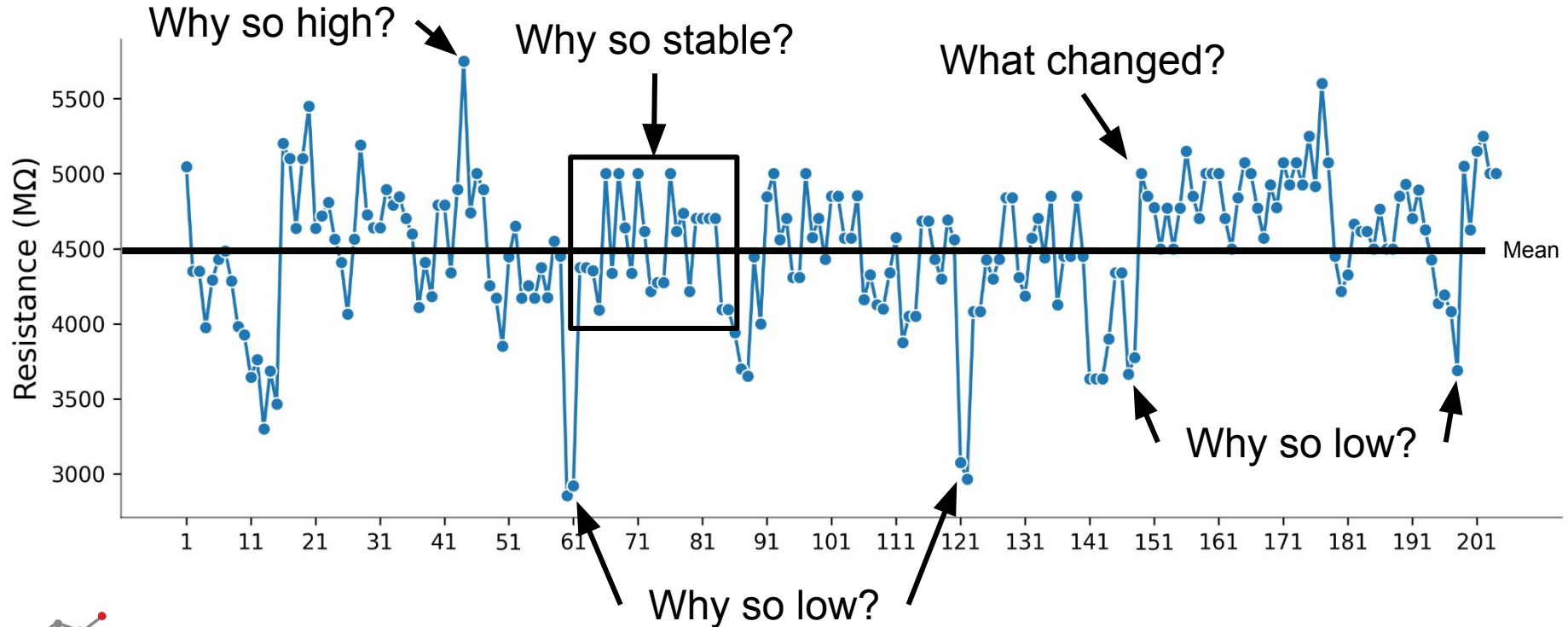
Standard deviation:
466.4



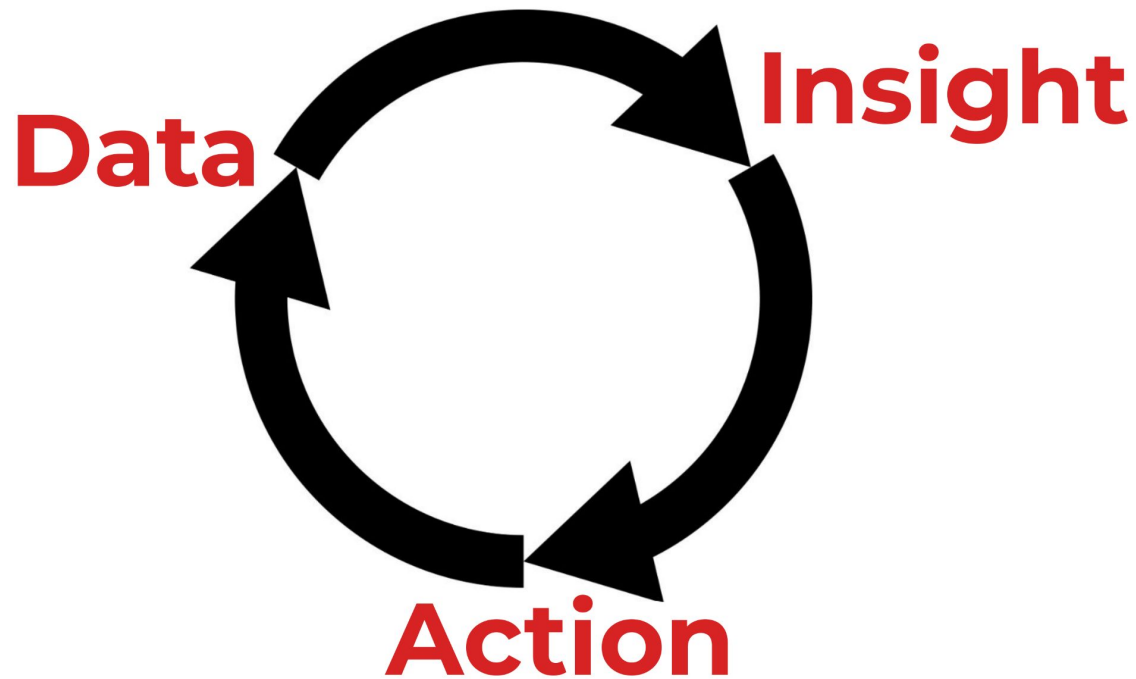
Given the histogram, what next?



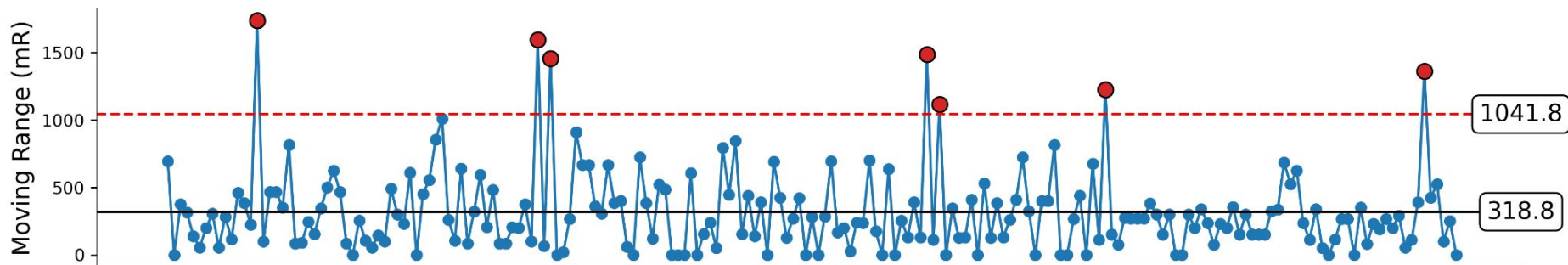
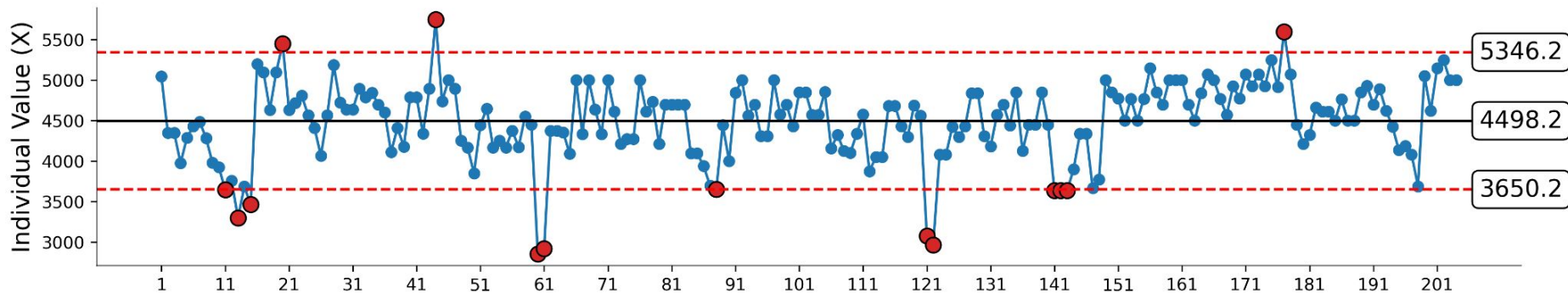
What about a time series?



The DIA Cycle



XmR chart of 204 resistance measurements



Shewhart's Generic Formula:

$$\text{Average} \pm 3 \cdot \text{Sigma}$$

XmR Chart Formulas:

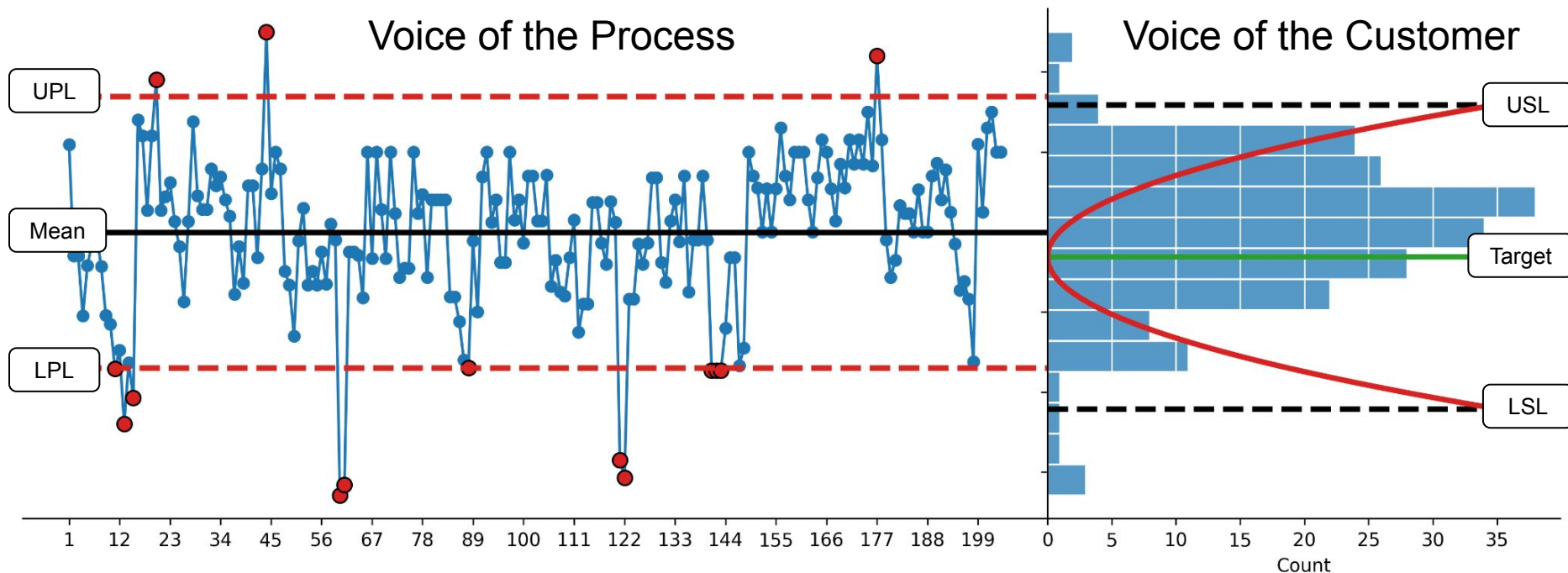
$$\text{UPL} = \text{Average} \pm (2.66 \cdot \text{Ave. mR})$$

$$\text{LPL} = \text{Average} \pm (2.66 \cdot \text{Ave. mR})$$

$$\text{URL} = 3.268 \cdot \text{Ave. mR}$$

Note: Sigma is NOT the standard deviation statistic. It is one of several within subgroup measures of dispersion that change with subgroup size.

Combo chart of resistance measurements



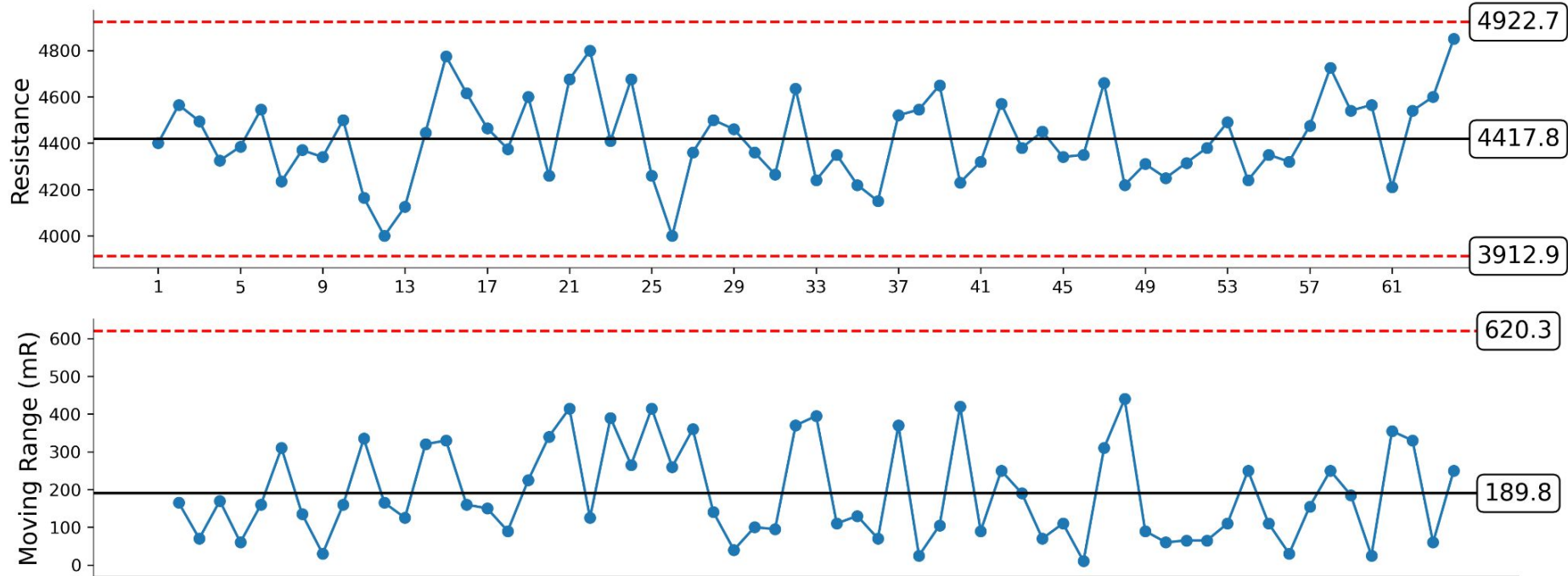
Further research was instituted...to find these causes of variability. Several were found, and after these had been eliminated another series of observed values gave the results indicated...Here we see that all points lie within the limits.

— Walter A. Shewhart
Economic Control of Quality of Manufactured Product

64 electrical resistance measurements

4,400	4,340	4,465	4,260	4,240	4,320	4,310	4,475
4,565	4,500	4,375	4,000	4,350	4,570	4,250	4,725
4,495	4,165	4,660	4,360	4,220	4,380	4,315	4,540
4,325	4,000	4,260	4,500	4,150	4,450	4,380	4,565
4,385	4,125	4,675	4,460	4,520	4,340	4,490	4,210
4,545	4,445	4,800	4,360	4,545	4,350	4,240	4,540
4,235	4,775	4,410	4,265	4,650	4,600	4,350	4,600
4,370	4,615	4,675	4,635	4,230	4,220	4,320	4,850

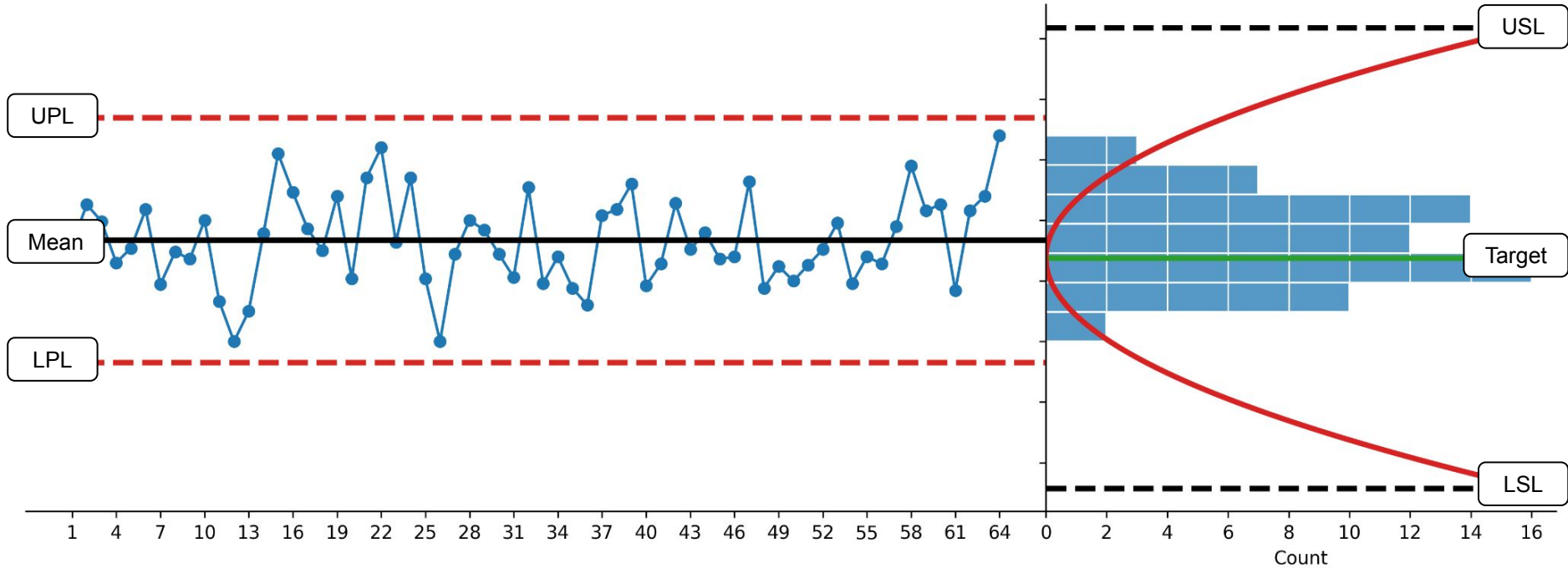
XmR chart of 64 resistance measurements



Combo chart of 64 resistance measurements

Voice of the Process

Voice of the Customer



Characterization:

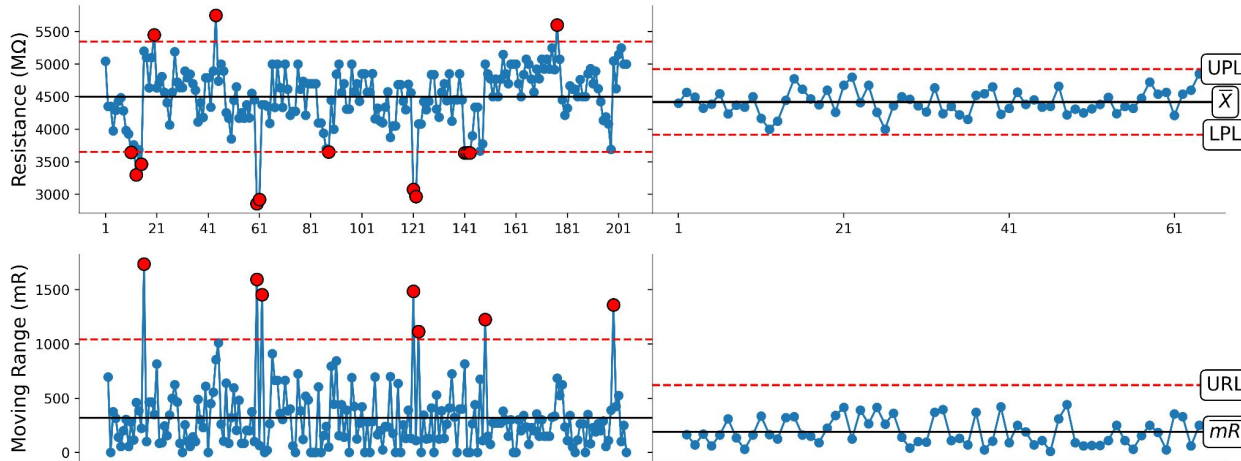
Is all the data
within the process limits?

No

Yes

Process is characterized as
unpredictable

Process is characterized as
predictable



The challenge of manufacturing is...

the challenge of reliably and repeatedly producing products that are, for all intents and purposes, the same.

To reliably and repeatedly produce products that
are the same...

we must know what kinds of variation influence
process behavior and respond in kind.

The only tool capable of this task is the
process behavior chart.

Questions

Additional resources

- A Crash Course in Variation (store.brokenquality.com/b/crash-course)
- Xmr Chart-It Web App (xmchart.streamlit.app)
- Taguchi Loss Web App (taguchi-loss.streamlit.app)
- The Broken Quality Initiative (brokenquality.com)
- The Broken Quality Initiative Store (store.brokenquality.com)
- The Math Matters Newsletter (themathmatters.substack.com)